



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Batch: A-2
Division: A

Subject: Programming with Java

ASSIGNMENT No : 21

TITLE : Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).

Learning Objectives

To understand and use the core JDBC classes for database connectivity.

Theory

The JDBC API consists of several classes and interfaces that facilitate database connectivity. The **DriverManager** class is responsible for loading the appropriate JDBC driver and establishing a connection with the database using a database URL, username, and password. The **Connection** interface represents an active session between the Java application and the database server. It provides methods to create SQL statements and manage transactions. The **Statement** interface executes static SQL queries and updates. These classes work together to establish communication between Java programs and databases. Understanding these core components is essential for developing database applications using Java.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp21/Main.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp21.Main
Database connected successfully.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

CODE 1

```
1 package Exp21;
2
3 import java.sql.*;
4
5 public class Main {
6     public static void main (String[] args) {
7
8         try {
9             Class.forName("com.mysql.cj.jdbc.Driver");
10
11             Connection con = DriverManager.getConnection(
12                 "jdbc:mysql://localhost:3306/college",
13                 "root",
14                 "devansh@2007"
15             );
16
17             Statement stmt = con.createStatement();
18
19             System.out.println("Database connected successfully.");
20
21             con.close();
22         }
23         catch (Exception e) {
24             System.out.println("Connection failed: " +
25 e.getMessage());
26         }
27 }
```

EXERCISE 1

```
1 package Exp21.Exercise;
2
3 import java.sql.*;
4
5 public class Exp1 {
6     public static void main(String[] args) {
7         try {
8             Class.forName("com.mysql.cj.jdbc.Driver");
9
10            Connection con = DriverManager.getConnection(
11                "jdbc:mysql://localhost:3306/college",
12                "root",
13                "devansh@2007"
14            );
15
16            if (con != null) {
17                System.out.println("Database connection
successful.");
18            }
19
20            con.close();
21        }
22        catch (Exception e) {
23            System.out.println("Database connection failed.");
24        }
25    }
26 }
```

EXERCISE 2

```
1 package Exp21.Exercise;
2
3 import java.sql.*;
4
5 public class Exp2 {
6     public static void main(String[] args) {
7         try {
8             Class.forName("com.mysql.cj.jdbc.Driver");
9
10            Connection con = DriverManager.getConnection(
11                "jdbc:mysql://localhost:3306/college",
12                "root",
13                "devansh@2007"
14            );
15
16            if (con != null) {
17                System.out.println(
18                    "Student database connected successfully."
19                );
20            }
21
22            con.close();
23        }
24        catch (Exception e) {
25            System.out.println("Student database connection
26failed.");
27        }
28 }
```

SQL CODE:

```
1 CREATE DATABASE college3;
2
3 USE college3;
```

Exercise

Q1. Create a program to connect Java with database and display database connection status.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp21/Exercise/Exp1.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp21.Exercise.Exp1
Database connection successful.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Create a Java program to connect with a database and display the connection status. After successful connection, show a message indicating that the Student database is connected successfully.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ javac -cp "/usr/share/java/mysql-connector-j.jar" Exp21/Exercise/Exp2.java
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java -cp ".:usr/share/java/mysql-connector-j.jar" Exp21.Exercise.Exp2
Student database connected successfully.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

Thus, the experiment successfully demonstrated the use of the core JDBC classes and interfaces for establishing communication between a Java application and a database. The roles of **DriverManager**, **Connection**, and **Statement** were understood for establishing a database connection and executing SQL statements. This experiment provided a basic understanding of how Java applications interact with databases using JDBC.